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PRACTICE BRIEFING

Real estate appraisal: a review of valuation methods

Elli Pagourtzi and Vassilis Assimakopoulos
School of Electrical and Computer Engineering, National Technical University of Athens, Athens, Greece
Thomas Hatzichristos
School of Rural and Surveying Engineering, National Technical University of Athens, Athens, Greece, and
Nick French
Jonathan Edwards Consulting, University of Reading, UK

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Abstract *The valuation of real estate is a central tenet for all businesses. Land and property are factors of production and, as with any other asset, the value of the land flows from the use to which it is put, and that in turn is dependent upon the demand (and supply) for the product that is produced. Valuation, in its simplest form, is the determination of the amount for which the property will transact on a particular date. However, there is a wide range of purposes for which valuations are required. These range from valuations for purchase and sale, transfer, tax assessment, expropriation, inheritance or estate settlement, investment and financing. The objective of the paper is to provide a brief overview of the methods used in real estate valuation. Valuation methods can be grouped as traditional and advanced. The traditional methods are regression models, comparable, cost, income, profit and contractor's method. The advanced methods are ANNs, hedonic pricing method, spatial analysis methods, fuzzy logic and ARIMA models.*

Introduction

Real property is defined as all the interests, benefits, rights and encumbrances inherent in the ownership of physical real estate, where real estate is the land together with all improvements that are permanently affixed to it and all appurtenances associated thereto.

The valuation of real estate is therefore required to provide a quantitive measure of the benefit and liabilities accruing from the ownership of the real estate. Valuations are required, and often carried out, by a number of different players in the marketplace. These may include:

- real estate agents;
- appraisers;
- assessors;
- mortgage lenders;
- brokers;
- property developers;

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- investors and fund managers;
- lenders;
- market researchers and analysts;
- shopping centre owners and operators; and
- other specialists and consultants.

This paper aims to examine the valuation of real estate prices, using prediction strategy based on selection of the best fitting model for use. The objective of the paper is to review the various methods used in real estate valuation.

The role of valuation

For any valuation to have validity it must produce an accurate estimate of the market price of the property. The model should therefore reflect the market culture and conditions at the time of the valuation. It should be remembered that the model should be a representation of the underlying fundamentals of the market. Thus, in the property market, what is often called a "valuation" is the best estimate of the trading price of the building.

In this context, the following convention is adopted:

- price is the actual exchange price in the marketplace;
- market value is an estimation of that price were the property to be sold in the market; and
- calculation of worth is used to assess the inherent worth to the individual or group of individuals.

In many property markets it is commonplace for the ownership of property to be separate from its use. Often the price of exchange will be the same whether the purchaser has investment or occupation in mind, but nonetheless the view of the two groups of bidders will be different. An investor will view worth as the discounted value of the rental stream produced by the asset, whereas the owner-occupier will see the asset as a factor of production and assign to it a worth derived from the property's contribution to the profits of the business. No doubt both groups of bidders will also be mindful of its potential resale price to a purchaser from the other group.

The concept of the worth of a property is most important in markets that are underdeveloped in terms of liquidity and the separation of ownership and use rights. Here most transactions are based on owner-occupiers' views of the worth of the property, i.e. the contribution it will make to business profit, as well as subjective issues such as status and feelings of security. Valuers, with hardly any transaction evidence, can only attempt to replicate these calculations of worth in arriving at an estimate of exchange price.

One of the paramount concerns of the valuation profession is the need to ensure that information presented to a client is clear and unambiguous. Not only should all parties understand the terminology used, it is also important

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that the client receives all other information that might be required to make a rational financial or investment decision. The latter point does not only concern the semantics of definitions of exchange price (see below), but must also address the issue of valuation methodology. Given that clients are themselves becoming more sophisticated in the way they determine whether to buy or sell property, then the pricing model used to assess the most likely exchange price should reflect their thought processes. This requires the valuer to better understand the client's requirements and leads to the adoption of more advanced valuation models which can reflect the increased level of data and information available.

Market value

A definition of value is an attempt to clarify the assumptions made in estimating the exchange price of a property if it were to be sold in the open market. These assumptions can include the nature of the legal interest, the physical condition of the building, the nature and timing of the market, and assumptions about possible purchasers in that market. Given that a compelling reason for using market value definitions is to ensure consistency in the process of valuation, it is important that there is a consistency of definition in all countries. For this reason, the International Valuation Standards Committee (IVSC) has set a "standard" to provide a common definition of market value. Market value is a representation of value in exchange, or the amount a property would bring if offered for sale in the open market at the date of valuation under circumstances that meet the requirements of the market value definition. In order to estimate market value, a valuer must first estimate the highest and best use, or most probable use. That use may be a continuation of a property's existing use or some alternative. These determinations are made from market evidence.

Market value is estimated through the application of valuation methods and procedures that reflect the nature of property and the circumstances under which the given property would most likely trade in the open market. Market value is defined for the purpose of the standards as follows:

> Market value is the estimated amount for which an asset should exchange on the date of valuation between a willing buyer and a willing seller in an arm's length transaction after proper marketing wherein the parties had each acted knowledgeably, prudently and without compulsion.

This paper reviews the various methods available to the valuer to estimate market value.

Methods

Each country will have a different culture and experience, which will determine the methods adopted for any particular valuation. The majority of all methods will rely upon some form of comparison to assess market value. This may be done, in its simplest form, by direct capital comparison or may rely upon a

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range of observations that allow the valuer to determine a regression model. Any such method is referred to in this paper as "traditional".

Other models or methods try to analyse the market by directly mimicking the thought processes of the players in the market in an attempt to estimate the point of exchange. These models tend to be more quantitive in method and will be referred to as "advanced".

For each method (or approach) that is described below, its theory is briefly explained together with an outline of how it is applied in the valuation process. The appropriate economic principles are also quoted with an explanation of how they apply to each method.

Methods can be grouped as follows:

(1) *Traditional valuation nethods*:

- comparable method;
- investment/income method;
- profit method;
- development/residual method;
- contractor's method/cost method;
- multiple regression method; and
- stepwise regression method.

(2) *Advanced valuation methods*:

- artificial neural networks (ANNs);
- hedonic pricing method;
- spatial analysis methods;
- fuzzy logic; and
- autoregressive integrated moving average (ARIMA).

Traditional valuation methods

Comparable method

Sales comparison is the most widely used approach. The value of the property being appraised (called the subject property) is assumed to relate closely to the selling prices of similar properties within the same market area. The appraiser first selects several similar properties (comparables or simply comps) from among all the properties that have recently been sold. Since no two properties are identical the appraiser must adjust the selling price of each comparable to account for differences between the subject and the comparable, i.e. differences in size, age, quality of construction, selling date, surrounding neighbourhood, etc. The appraiser infers the current value of the subject from the adjusted sales prices of the comparables.

The sales comparison approach is heavily dependent on the availability, accuracy, completeness, and timeliness of sale transaction data. Information

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The objective of the paper is to survey the functional forms (methods) used in real estate estimation. In this way, we can use the appropriate method according our criteria to estimate property value. There is continuing debate about the interpretation of value concepts by means of definitions of value and their implementation by means of a valuation methodology. As valuers move from operating in their home country to the demands of a European and international marketplace, these issues are likely to become more complex. Conversely, the cross fertilisation of ideas provides an opportunity for improved theory and practice.

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sources include government records, data vendors, and the appraiser's network of local contacts (e.g. brokers participating in transactions (Castle and Gilbert, 1998)).

Comparable sales analysis procedure may be viewed as a four-part process:

(1) For a given subject property, finding the most comparable sales.

(2) Adjusting the selling prices of the comparables to match the characteristics of the subject.

(3) Using the several estimates of value to arrive at an estimate of market value.

(4) Presenting the results in a report format suitable for viewing or printing.

The process of finding comparables utilizes "distance" to establish a measure of comparability between the subject and the comparable under consideration. It is computed by weighting the differences in characteristics between the subject and the comparable. The distance, D is calculated as follows (McCluskey *et al.*, 1997):

$$D = \sqrt[\lambda]{\sum_i [A_i(X_i - X_{si})]^\lambda + \sum_j [A_j \bar{\delta}(X_j, X_{sj})]^\lambda}, \tag{3}$$

where:

λ = Minkowski exponent lambda
A_i = weight associated with the ith continuous characteristic
X_i = value of the ith characteristic in the sale property
X_{si} = value of ith characteristic in subject property
$\sum_i$ = summation of terms of i characteristics
A_j = weight associated with the jth categorical characteristic
X_j = value of jth characteristic in sale property
X_{sj} = value of jth characteristic in subject property
$\sum_j$ = summation of terms of j characteristics
$\delta(a, b)$ = inverse delta function (0, if $a = b$; 1, if $a \neq \delta b$).

For each comparable property the sales price is adjusted to the subject property as follows:

Adjusted sales price = sales price – (comparable MRA – subject MRA).

Given the several comparable sales, several adjusted selling prices are obtained. A weighted estimate is formed as follows:

$$\text{Weighted estimate} = \sum_{i=1}^{n} \frac{W_i}{W} ASP_i, \tag{4}$$

where the weight for the comparable is:

$$i = W_i = \frac{1}{(D/2)^2 + D_j^2 + [2D(|ASP_i - SP_i|/SP_i)]^2}, \tag{5}$$

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$$W = \sum_{i=1}^{n} W_i, \tag{6}$$

where:

ASP_i = adjusted sale price for comparable i

SP_i = sale price of comparable I

D_i = distance for comparable I

D = max of D_i.

Thus the weighted estimate of value places more emphasis on properties, which are most like (smaller distances) the subject property and have the smaller adjustments to the selling price. The comparables with the lowest distance are selected. In calculating the distance the variables are allocated a factor weight (McCluskey and Borst, 1997). The weighting is used to balance the effect of variables according to the magnitude of the variable itself, so that a variable with larger numerical size has a smaller weight. This process of computing several comparable sales estimates of value, a weighted estimate of value and an MRA (multi regression analysis) estimate of value yields, in the case of five comparable sales, seven estimates of value.

Investment/income capitalisation method

At its simplest level, the comparable model can be used to determine capital value directly. However, moving from sub-markets where there is a high degree of similarity (for example, residential markets), the way in which comparison can be utilised needs to be modified.

In the investment market, for example, direct capital comparison is rarely appropriate because the degree of heterogeneity is much higher. As such, the comparison needs to be broken down further to look at rental (on a *pro-rata* basis) and the initial yield achieved on sale. This distinction between the rental and the yield reflects an interesting interaction between two sub-markets, the occupational market and the investment market. At its simplest level, property can either be owned and occupied by the same party (owner-occupied), or the owner can choose to pass the right of occupation to a third party by letting the property. The tenant will then pay the owner (the landlord) a rent to represent the (normal) annual value of the property to the tenant. The level of rent is determined by the supply of, and demand for, that type of property in the market. The rent also represents the return or interest on the money invested in the property by the owner. It is the remuneration for the giving up of the use of the property. This rental income is simply a cash flow and as such the value of the rented property may be determined by the present value of the predicted cash flow.

Similarly, it is possible to determine a gross rent multiplier by analysing other previous sales. Investors can be determined to be paying "x" times the rent for a particular type of property. The higher the multiplier the higher the

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market value, and this in turn reflects the greater attractiveness of the subject property. Thus, valuers decapitalise recent comparable sales and apply the derived multiplier to the rent of the subject property. The investment method is a method of simple comparison. It does not attempt to analyse the worth of the property investment from first principles.

Profits method

In the section above, the investment method is shown to be a method where valuation methodology has moved away from modelling the thought processes of the players in the market, and instead assesses the market value of a subject property by reference to observed recent transactions of similar properties in the same area. It no longer looks at the fundamentals; the original reasons why the purchasers might be willing to buy at a certain price for such assets in the market.

However, if there are insufficient sales to determine a comparable value and if there is no rent produced because the property is in owner-occupation, then the valuer must determine the value by returning to a detailed market analysis. For instance, the market value of a hotel in owner-occupation will be dependent on the potential cash flow to be derived from ownership. That cash flow will be determined by the number of bedrooms in the hotel, the room rate and the average occupancy rate for the year. In other words, property is simply viewed as a unit of production and it is the valuer's role to assess the economic rent for the property from first principles. This is calculated by assessing the potential revenue to be expected each year from the hotel, and deducting all other costs of a prudent hotelier in realising that cash flow. These costs will include direct costs such as catering, laundry and service. In addition, allowances will need to be made for the remuneration of the hotelier, interest on money borrowed to run the hotel and a return on capital for any equity tied up in the business. Having calculated the liabilities these are deducted from the revenue figure and the residue will be an estimation of the economic rent for the property. The capital value can then be derived by multiplying the annual rent by an appropriate multiplier.

This process reverts to a fundamental analysis of the worth of the property to the business. The economic rent is a derivative of the supply and demand for the final product, in this example, the hotel rooms. The same principle will apply to any type of property where the market value of the property is intrinsically linked to the business carried out within that property. Other examples will therefore include restaurants, leisure centres, cinemas, theatres, etc.

Development/residual method

The properties under estimation are plots or sites that can be developed. The best method for estimating site value is through comparable vacant land sales. The sales should be reduced to appropriate units of comparison. The value of the land or site should be estimated as if the site were vacant and available for

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its highest and best use. Each comparable sale should be described. As a minimum, the description must include the following data:

- location;
- grantor;
- grantee;
- recording data;
- date;
- sale price;
- financing;
- units of comparison;
- lot dimensions;
- configuration and size;
- physical and topographical characteristics;
- zoning, utilities; and
- environmental influences.

This analysis of understanding the market value of the land and property to the business can be extended to include the valuation of development property. If one views the process of (re)development as a business, it is possible to assess the market value of land and buildings in their existing form as part of that process. Development occurs where the current use of land and buildings is not the highest and best. By spending money redeveloping the site, it is possible to release latent value, as the market value of the land is increased due to the demand for the new use commanding a higher price than the previous use. By viewing development in this way, it can be seen that the residual method of valuation is very similar to the profits method.

With the residual method, the valuer assesses the market value of the land in a redeveloped form (either by comparison or by the investment method) and deducts from this gross development value all costs that will be incurred in putting the property into the form that will command that price. These costs will include demolition of the existing building (if not already a cleared site), infrastructure works, construction costs, professional fees, finance costs and a remuneration for undertaking the risk of development (developer's profit). By deducting these liabilities from the final market value, a residue is produced. This residue represents the maximum capital expenditure for buying the land. It will therefore include all costs of purchase (taxation, legal fees, professional fees and finance). The net residual land value is determined by allowing for these additional land costs. It can be seen therefore that the residual land value is, as with any economic rent, dependent upon the supply and demand of the finished product, the developed property. The greater the demand for the finished property, the higher the gross development value,

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and if costs remain relatively static, the higher the market value of the land in its original state.

Contractor's/cost method

A further way in which it is possible to estimate the market value of land and property is the contractor's method or the replacement cost method. If the property being valued is so specialised that properties of that nature are rarely sold on the open market, it will be effectively impossible to assess its value by reference to comparable sales of similar properties. Similarly, if there is no rental produced, the investment method will also be inappropriate. The profits method could be applied if the property is intrinsically linked to the business carried out in the property, however, where that business is one of production (rather than service) it is difficult to determine the contribution of the property to the overall usage. The plant and machinery contained within are likely to have a greater value to the business than the structure containing them. Thus, once again, the valuer must revert to understanding the thought process of the user of the building. This can be illustrated by reference to a property such as an oil refinery. Here the nature of the business is so specialised that there are no comparisons, the property would be owner-occupied so there is no rental and the plant and the machinery will be the important elements contributing to the value of the business. Thus, the owner of the building will simply assess the market value of the building by reference to its replacement cost. How much would it cost to replace the property, if the business were deprived of its use? In simple terms, market value will equate to reconstruction costs. The valuer will assess the market value of the raw land (by reference to comparable land values in an appropriate alternative use), add to this value the cost of rebuilding a new building which could perform the function of the existing structure, and from this then make subjective adjustments to allow for the obsolescence and depreciation of the existing building relative to the new hypothetical unit. It is reasonable to assume that this mirrors the thought process of the owner-occupier and thus should be viewed as a valid and rational method of valuation.

It is interesting that in countries where property investment is less prevalent and where owner-occupation is the favoured method of property utilisation, then it is not only specialised properties which are valued by the contractor's method. If there is no investment market (i.e. properties will only exchange between owner-occupiers in the market) then the price of exchange will reflect the "bottom line" cost to the purchaser. This bottom line will be the cost that will need to be incurred for a new build relative to the existing property that is on the market. There will be a strong correlation between price and cost. However, if the occupation market is dominated by companies renting, and there is a degree of scarcity in the market, then price will be determined not by cost, but by the supply and demand characteristics of the occupational market. In such a case, regardless of the nature of the property, the investment method will dominate as the favoured valuation model.

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Multiple regression method

The general multiple linear regression models is:

$$_i = \beta_0 + \beta_1 X_{1,i} + \ldots + \beta_k X_{k,i} + \epsilon_I, \quad (1)$$

where $Y_i, X_{1,i}, \ldots, X_{k,i}$ represent the ith observations of each of the variables $Y_i, X_1, \ldots, X_k$ respectively, $\beta_0, \beta_1, \ldots, \beta_k$ are fixed (but unknown) parameters and ϵ_i is a random variable that is normally distributed with mean zero and having a variance σ_ϵ^2.

There are several assumptions made about X_i and ϵ_i which are important:

- The explanatory variables $X_1, \ldots, X_k$ take values which are assumed to be either fixed numbers (measured without error), or they are random but uncorrelated with the error terms ϵ_i. In either case, the values of $Xj(j = 1, 2, \ldots, k)$ must not be all the same.
- The error terms ϵ_i are uncorrelated with one another.
- The error terms ϵ_i all have mean zero and variance σ_ϵ^2, and have a normal distribution (Makridakis *et al.*, 1998).

There is an example of regression analysis in real estate (Wolverton, 1997). The data for this example consisted of 56 residential, mountainside view lots located in Tucson, Arizona, on sale over the 1989-1991 period. The data are restricted to a relatively small geographic area to control for variation in household income and other exogenous price influences. All the sale properties are located within the same public school district, subject to the same governmental jurisdiction and property tax rates, and are equally distant from major employment nodes.

The characteristic variables of the model are:

- quality of city view (VIEW) was measured by metrically scaling the width of each lot's angle of city view panorama, adjusted for blockage or potential blockage from nearby homes;
- lot size (SIZE) was taken from recorded plots;
- a dummy variable (DEV) was coded as one for seven of the lots in the data set involved steep terrain and consequently high-expected expenditures for site fill and building foundations, and zero otherwise; and
- variables that describe 21 sales which occurred in 1988, 11 in 1989, 19 in 1990, and five in 1991.

The example of the functional form of diminishing marginal price deals with the price per square foot and the second subsection deals with the price per degree of included angle of view. As a first step in the analysis of diminishing marginal price per square foot, lot price per square foot (PSF) was regressed on lot developability (DEV), lot size (SIZE) measured in thousands of square feet, city view (VIEW), and year of sale (YR 1989, YR 1990 and YR 1991) with 1988

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as the base year (referred to herein as the "naïve" model). This first regression model demonstrates that these variables account for most of the variability in lot price per square foot (adj. $R^2 = 0.77$), and that DEV, VIEW and SIZE are all significant determinants of price in this sub-market, with p values of 0.001 or better.

The estimation model is depicted by:

$$P_{SF} = \beta_0 + \beta_i(DEV) + \beta_2(VIEW) + \beta_3(YR1989) + \beta_4(YR1990) + \beta_5(YR1991) + \delta. \quad (2)$$

Stepwise regression method

Stepwise regression is a method which can be used to help sort out the relevant explanatory variables from a set of candidate explanatory variables when the number of explanatory variables is too large to allow all possible regression models to be computed.

One of the main kinds of stepwise regression in use today is called "stepwise forward-with- a-backward-look regression" and is explained below:

- *Step 1*: Find the best single variable (X_{1*}).
- *Step 2*: Find the best pair of variables (Y_{1*} together with one of the remaining explanatory variables – call it X_{2*}).
- *Step 3*: Find the best triple of explanatory variables (X_{1*}, X_{2*} plus one of the remaining explanatory variables – call the new one X_{3*}).
- *Step 4*: From this step on, the procedure checks to see if any of the earlier introduced variables might conceivably have to be removed. For example, the regression of Y on X_{2*} and X_{3*} might give better R^2 results than if all three variables X_{1*}, X_{2*} and X_{3*} had been included. At step 2, the best pair of explanatory variables had to include X_{1*}, by step 3, X_{2*} and X_{3*} could actually be superior to all three variables.
- *Step 5*: The process of looking for the next best explanatory variable to include, and checking to see if a previously included variable should be removed, is continued until certain criteria are satisfied. For example, in running a stepwise regression program, the user is asked to enter two "tail" probabilities:

  the probability, P_1, to "enter" a variable; and

  the probability, P_2, to "remove" a variable.

When it is no longer possible to find any new variable that contributes at the P_1 level to the R^2 value, or if no variable needs to be removed at the P_2 level, then the iterative procedure stops (Makridakis *et al.*, 1998).

An example about stepwise regression is presented. In the current example, a model is performed on some 2,405 cottages sold in the Quebec Urban Community (QUC) from January 1993 to January 1997. Cottages present the

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advantage of being spread all over the QUC, as opposed to bungalows (one-story, detached, houses) and to condominium units, found mainly in suburban areas in the former case and in central neighbourhoods in the latter. Sale prices of sampled cottages range from $50,000 to $250,000, with mean price standing at $123,183. Many attributes are available to describe these transactions. They can be grouped as follows:

- transaction attributes (mainly sale price, the dependent variable);
- property specifics (66 attributes in the initial data set; 22 selected during stepwise regression analysis – models A to D);
- local taxation attributes (two available; two selected by the model – models A to D);
- neighbourhood attributes (34 relative attributes – models C and D);
- proximity attributes designed at capturing externalities (19 initial variables – models B to D); and
- travel accessibility measured on the street network (15 provided – model D).

Following a five-step approach, property specifics are first introduced in the model; proximity and neighbourhood attributes are then successively added on. Finally, factor analyses are performed on each set of access and census variables, thereby reducing to six principal components an array of 49 individual attributes. Substituting the resulting factors for the initial descriptors leads to high model performances, and controlled colinearity, although remaining spatial autocorrelation is still detected in the residuals (Des Rosiers *et al.*, 2000).

Advanced valuation methods

Artificial neural networks (ANNs)

Artificial neural network models have been offered as a possible solution to many problems in real estate valuation. An artificial neural network model must first be trained from a set of data and the model is then utilized to estimate the prices of new properties from the same market. Neural networks are artificial intelligence models originally designed to replicate the human brain's learning processes.

These models have three primary components:

(1) the input data layer;

(2) the hidden layer(s), commonly referred to as the "black box"; and

(3) the output measure(s) layer, the estimated property value(s).

The hidden layer(s) contain two processes: the weighted summation functions; and the transformation functions. Both of these functions relate the values from the input data (e.g. the property attributes: number of bathrooms; age of house; lot size; basement area; total area; number of fireplaces; number of garages) to

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the output measures (the sales price). The weighted summation function typically used in a feed-forward/back propagation neural network model is:

$$Y_j = \sum_{j}^{n} X_i W_{ij}, \tag{7}$$

where X_i is the input values and W_{ij} is the weights assigned to the input values for each of the j hidden layer nodes. A transformation function then relates the summation value(s) of the hidden layer(s) to the output variable value(s) or Y_j. This transformation function can be of many different forms: linear functions, linear threshold functions, step linear functions, sigmoid functions or Gaussian functions. Most software products utilize a regular sigmoid transformation function such as:

$$Y_T = \frac{1}{1 + e^{-y}}. \tag{8}$$

This function is preferred due to its non-linearity, continuity, monotonicity, and continual differentiability properties (Borst, 1992; Trippi and Turban, 1993).

There is research about three artificial neural networks for estimating the value of a random sample of "normal" residential properties and a sample of outlier properties. The data used in this research consist of 288 single-family residential properties that were sold in Fort Collins, Colorado, USA from November 1993 to January 1994.

The variables that determine value were the number of bathrooms, the age of the house, the lot size, the finished interior square footage of the house, whether there was a basement, the number of fireplaces, and the size of the garage. The log of the property sales price was used as the output layer for the artificial neural network model.

Outlier properties were determined as properties that possessed a z-score greater than 2.0. A z-score was measured by subtracting the property price from the average price of the houses in the sample and dividing by the sample standard deviation. A total of 17 outlier properties were identified and separated into an "outlier" holdout sample, leaving 271 properties in the "normal properties" data set. The remaining 271 properties were sorted by price and every fourth property was separated out into a "normal" holdout sample, leaving 204 properties to be the training sample for creating the artificial neural networks.

The model with the optimal number of hidden layer nodes would possess the minimum mean absolute prediction error and the maximum number of houses within a 5 per cent absolute prediction error of the actual sales price. Six hidden layer nodes were found to be the optimal number of nodes within the hidden layer for the three artificial neural networks.

Hedonic pricing models

The theory of hedonic price functions provides a framework for the analysis of differentiated goods like housing units, whose individual features do not have

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observable market prices. A differentiated product can be represented as a vector of characteristics with the market price dependent upon the set of features. The market price of the product implicitly reveals the hedonic price function relating characteristics to prices. The traditional use of hedonic estimation in housing studies has been for the purpose of making inferences about non-observable values of different attributes like air quality, airport noise, commuter access (railway, subway or highway) and neighbourhood amenities (Janssen *et al.*, 2001).

Spatial analysis methods

While GIS can improve the measurement of location and access variables, namely by resorting to time, rather than mere Euclidean distances, their analytical capabilities are greatly enhanced where spatial statistics methods are integrated (Anselin and Getis, 1992; Griffith, 1993; Zhang and Griffith, 1993; Theriault and Des Rosiers, 1995; Levine, 1996). Indeed, procedures such as spatial pattern analysis and autocorrelation analysis (Odland, 1988; Cressie, 1993; Ord and Getis, 1995; Tiefelsdorf and Boots, 1997) as well as variography and Kriging techniques (Dubin, 1992; Panatier, 1996) can help detecting additional neighbourhood factors that must be considered to explain market variability, as it is described below.

An alternative to the establishment of fixed neighbourhoods or composite sub markets involves a more rigorous spatial analysis of property prices in terms of developing terrain of surface models using spatial interpolation. The spatial interpolation techniques use a set of data based on discrete points for sub-areas, then determine a function that will best represent the whole surface which can then be used to predict values at other points or sub-areas. Chou (1997) states the two fundamental assumptions underlying spatial interpolation. First, the surface of the z-variable is continuous; therefore the data value at any location can be estimated if sufficient information about the surface is given. Second, an implicit assumption is that the z-variable is spatially dependent; in other words the interpolation of the variable value can be extracted from the given spatial distribution because the value at any specific location is related to the values of surrounding locations. Numerous algorithms for point interpolation have been developed; however, the selection of the appropriate algorithm depends largely on the type of data, the degree of accuracy required and the amount of computational effort (Lam, 1983). According to Lam (1983) the point interpolation methods can be classified as either exact or approximate methods.

Within a GIS framework the use of surface response analysis techniques has been shown to provide a three-dimensional visualization of the value of location as it varies geographically. While research in this area has been limited, the work which has been carried out has contributed to a better understanding of the measurement of location effects. Research into triangulated irregular networks (TINs) by LaRose (1988) demonstrated the potential of this approach to predictive modelling. This research was also interesting from the perspective

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that a global TIN produced better results that a TIN which was based on a stratified subset. In further developing this line of research Des Rosiers and Theriault (1992) investigated the use of isovalue plots and three-dimensional models; Wyatt (1995) utilized three-dimensional images integrated with network models; Gallimore *et al.* (1996) investigated the use of MRA generated residuals (with no location variables) in building a response surface which could then be used to adjust for the under- or over-valuation of the property. A perceived problem with this approach is that the adjustment factor used was based on the error for each property; while this worked well with properties having known sale prices, its validity would need to be tested in relation to unsold properties.

Fuzzy logic

Classic Boolean logic is binary, that is a certain element is true or false, an object belongs to a set or it does not. Fuzzy logic, introduced by Zadeh in 1965, permits the notion of nuance. The key to Zadeh's idea is to represent the similarity a point shares with each group with a function (termed the membership function) whose values (called memberships) are between $0 < m < 1$. Each point will have a membership in every group, memberships close to unity signify a high degree of similarity between the point and a group, while membership close to zero implies little similarity between the point and that group. Additionally, the sum of the memberships for each point must be unity.

Every continuous math function can be approximated by a fuzzy set.

Several types of membership functions can be utilized. The membership function reflects the knowledge for the specific object or event.

Another critical step in the fuzzy systems methodological approach is the definition of the rules, which connect the input with the output. These rules are based on the form "if ... then ... and". The knowledge in a problem-solving area can be represented by a number of rules. For example, if the output set "value: is comprised by two subsets called: 'low' and 'high'," two rules could be:

(1) If the distance is small then value is low.

(2) If the distance is great then value is high.

In order to solve a problem with a knowledge-based fuzzy system it is necessary to describe and process the influencing factors in fuzzy terms and provide the result of this processing in a usable form. The basic elements of a knowledge-based fuzzy system are:

- fuzzification;
- knowledge base;
- processing; and
- defuzzification.

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The use of fuzzy logic for the analysis and the modelling of real estate could be a powerful tool in modern planning, as is pointed out by many researchers (Bagnoli and Smith, 1998; Gold, 1995; Byrne, 1994). The most important advantages of fuzzy modelling are:

- It is a more realistic approach through the use of linguistic variables instead of numbers.
- Hierarchical ranking of the objects (e.g. buildings, lots) and not an inclusion – exclusion list.
- Fewer repetitions of the model.

Autoregressive integrated moving average (ARIMA)

Autoregressive (AR) models can be effectively coupled with moving average (MA) models to form a general and useful class of time series models called autoregressive moving average (ARMA) models. However, they can only be used when the data are stationary. This class of models can be extended to non-stationary series by allowing differencing of the data series. These are called autoregressive integrated moving average (ARIMA) models. Box and Jenkins (1970) popularized ARIMA models.

There are a huge variety of ARIMA models. The general non-seasonal model is known as ARIMA (p, d, q):

AR: p = order of the autoregressive part.

I: d = degree of first differencing involved.

MA: q = order of the moving average part (Makridakis *et al.*, 1998).

There is some research that applies the Box-Jenkins methodology – ARIMA model to the study of Hong Kong's real estate prices (Tse, 1997). This example shows how the office and industrial property prices in Hong Kong can be fitted into the ARIMA equation. They used quarterly data in the period 1980Q1-1995Q2 (the year is separated in four quarters Q1, Q2, Q3, Q4) which contained 59 observations. The estimated equations have been used to forecast for the next three quarters. It is in any case difficult to identify the most appropriate proxy for the price index in the real estate market, since this heterogeneous sector includes different types and classes of building, and demand for them is generated across all sectors of the economy. Above all Hong Kong's properties are more homogeneous since multi-storey development has remained predominant in the real estate market.

The general form of the ARIMA model can be written as:

$$\phi(B)(1-B)^d Y_t = \theta(B)_{\epsilon_t}. \tag{9}$$

where B represents the back shift operator such that $BY_t = Y_{t+1}$, Y_t is the value of the time series observation at time t, ϵ_t is a series of random shocks which are assumed to be independently, normally distributed with zero mean and variance and d represents the order of difference. If a series is stationary,

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then $d = 0$. In equation (10), $\phi(B)$ is a polynomial of order p in the back shift operator B, which is defined as:

$$\phi(B) = 1 - \sum_{i=1}^{p} \phi_i B^i. \tag{10}$$

Similarly, $\phi(B)$ is defined to be a polynomial of order q in B, such that:

$$\theta(B) = 1 - \sum_{i=1}^{q} \theta_i B^i. \tag{11}$$

This is the only valuation method that depends on time variables. The ARIMA model is essentially an approach to economic forecasting based on time-series data (Dickey and Fuller, 1981; Granger and Newbold, 1974; Tse, 1996).

Conclusions

In this paper we have reviewed the methods that have been used for estimating real estate property's value. The existing European (UK) and North American (US) literature considers that the comparable method is accurate and reliable estimated method. Many researchers have their reservations about method's reliability because of the subjectivity of the key variable choice. In cases where there is lack of data we can use the comparable method. The surveyor imprints the property market in order to estimate the value of the property. He or she has to determine the comparative set of properties and recognizes the key variables. This method allows us to focus on selection, evaluation and registration of the value elements that is very important in appraisal.

Other methodologies that are also presented in this paper can resolve the problem of estimating the value of properties as a possibility in this regard. For example, the resulting regression coefficients provide estimates of the value of individual property features. This offers a scientific basis for the price adjustments and does not rely on the judgment and experience, or inexperience, of the appraiser or agent. Regression analysis can also handle many more comparables than the few generally used in comparative market analyses performed by real estate agents or accredited appraisers.

A dilemma in social science is that one often does not know which the appropriate model is. The procedure then is to reason through the issues, consult the literature, consider alternatives, choose a model, perform the analysis, and study the results. If the results do not give cause to refute the model, appear reasonable and logical, and in agreement with accepted beliefs, the model is regarded as appropriate. We proceed according these principles, paying particular attention to the following two issues (Janssen *et al.*, 2001):

(1) functional form; and

(2) variable selection.

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